

“In our language”: Acceptability and impact of executive function videos for Spanish-speaking families

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ABSTRACT

Purpose: Predominantly Spanish-speaking Hispanic/Latine families of children with neurodevelopmental conditions (e.g., autism, ADHD) in the U.S. face barriers accessing intervention services. This study evaluated the acceptability and effectiveness of executive function (EF) support videos translated into Spanish for caregivers in this community.

Methods: In this pragmatic trial, 37 caregivers of neurodivergent children aged 8–11 reviewed 12 brief EF-focused videos addressing skills like flexible thinking, planning, and self-regulation, with strategies for home use. The original English-language videos were developed with input from a Community Advisory Board (CAB) to ensure broad applicability across socioeconomic and racial/ethnic backgrounds, then directly translated and dubbed by a bilingual team. Caregivers completed pre/post measures assessing EF knowledge, caregiver strain, child EF difficulties, and video acceptability and feasibility. Focus groups were used to contextualize and expand on quantitative findings. An exploratory comparison was made to a prior English-language video cohort.

Results: Caregivers in the Spanish video cohort reported high acceptability and feasibility and demonstrated improvements in EF knowledge, reduced strain, and improved child EF outcomes. These outcomes were greater than those in the English video cohort. Focus group participants emphasized the relevance of the content, the practicality of the format, and the importance of receiving resources in their native language.

Conclusion: These findings highlight the potential of pragmatic, scalable, and linguistically accessible intervention resources for underserved communities. The EF videos, though directly translated with limited culturally specific adaptation, were well-received and effective in improving outcomes for Spanish-speaking caregivers.

Rates of Hispanic/Latine¹ individuals diagnosed with neurodivergent conditions, including autism and ADHD, have increased

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¹ The term Hispanic is used throughout the remainder of this manuscript as it is the preferred term of the majority of members of this community (Noe-Bustamante et al., 2024)

greatly in recent years (Chung et al., 2019; Maenner et al., 2023), leading to increased demand for intervention services. However, disparities exist in the availability and utilization of these services for Hispanic children with neurodivergent conditions (Alvarado & Modesto-Lowe, 2017; Smith et al., 2020). In the U.S., these families face significant challenges in accessing services due to factors such as language barriers, limited awareness of the different types of services, and logistical barriers associated with socioeconomic status (SES), with monolingual Spanish-speaking caregivers experiencing the most difficulty (Zuckerman et al., 2017). Similar challenges exist in Latin America, with one study finding that 76 % of caregivers in the region experienced barriers to accessing services for their autistic children during elementary school ages (Montenegro et al., 2022). Given that there are approximately 500 million native Spanish-speakers worldwide (Instituto Cervantes, 2021), it is critical to ensure there are sufficient resources available in the language.

A promising practice to reduce barriers to accessing treatment services is the use of caregiver telehealth interventions, which deliver support remotely via electronic information and telecommunication technologies such as websites, video calls, and mobile apps (HRSA, 2023). These types of interventions broadly fall into two categories, caregiver coaching interventions, which involve live coaching and real-time interaction with a professional, and caregiver trainings, which are typically asynchronous and self-paced, allowing caregivers to learn through pre-recorded videos or online modules. Both models aim to increase caregiver knowledge and efficacy while also reducing caregiver strain or sense of burden. Coaching models allow caregivers to practice strategies in real time with their children, which helps them apply techniques more effectively and supports the generalization of skills across settings and situations (Nevill et al. 2018). There is also evidence that these programs can be effectively delivered via telehealth (e.g., Pickard et al., 2016). However, these types of interventions require a trained clinician and can have similar barriers to access as in-person interventions for lower SES families or those who do not speak English (e.g., high cost, insurance requirements, inflexible service delivery hours, lack of linguistically matched clinician). In contrast, caregiver training offered online asynchronously can support caregivers of neurodivergent children without the need for a trained clinician. Several prior studies have shown that these types of trainings can increase caregiver knowledge of concepts and strategies to support their children (Dai et al., 2021; Kenworthy et al., 2023) and reduce their parenting-related strain (Dickter et al., 2021). However, all of the studies evaluating such asynchronous supports were conducted with English-speaking families.

Despite the increasing efforts to develop telehealth interventions, few have been intentionally designed for Spanish-speaking Hispanic caregivers of neurodivergent children. Existing research highlights both the promise and limitations of current approaches. For example, in some therapist-led programs, predominantly Spanish-speaking caregivers have reported high satisfaction with psychoeducational content, though have expressed mixed preferences regarding online versus in-person delivery (Safer-Lichtenstein et al., 2023). Other interventions targeting more specific skills, i.e. behavioral parent training or acceptance and commitment therapy, have also been successfully adapted for telehealth with this population, but have also predominantly been therapist led and with small samples (e.g. Baires et al., 2023; Vargas Londono et al. 2024). Only one study to date has explored the development of culturally adapted asynchronous online materials for Spanish-speaking caregivers (Buzhardt et al. 2016), but the materials were not evaluated for effectiveness. Taken together, these studies point to a growing foundation, but also a clear gap, in evidence-based asynchronous materials for Spanish-speaking caregivers of neurodivergent children.

An area consistently shown to be one of difficulty for neurodivergent children across several diagnoses, and thus a prime target for intervention, is executive functioning (EF) skills (Craig et al., 2016). EF encompasses brain-based skills such as flexible thinking, problem-solving, self-regulation, planning, and organization (Miyake et al., 2000; Stuss & Alexander, 2000). These skills are crucial for regulating thoughts, behaviors, and emotions to achieve goals and complete tasks, impacting lifelong outcomes in adaptive functioning, mental health, and school performance (Kenworthy et al., 2008; Lawson et al., 2015; Pugliese et al., 2016). Interventions aimed at improving EF skills are generally effective, with metanalytic studies showing moderate-to-large effect size improvements for non-pharmacological EF interventions for both autistic children (Kaur et al., 2024) and those with ADHD (Qiu et al., 2023).

One of the most widely studied non-pharmacological EF interventions is *Unstuck and On Target* (Unstuck, 2024; Cannon et al., 2021), a manualized cognitive-behavioral therapy shown to be effective for 8–11-year-old children with autism and/or ADHD in schools and clinics (Anthony et al., 2020; Elias et al., 2019; Kenworthy et al., 2014). *Unstuck* helps improve EF deficits by explicitly teaching children skills such as flexible thinking, emotional regulation, and goal-directed planning through self-regulatory scripts and direct practice, while also providing the adults in their lives (i.e., caregivers and teachers) tools to support development of these skills. In a school-based trial demonstrating *Unstuck*'s effectiveness with a culturally and linguistically diverse population, caregivers were offered four 90-minute supplemental information sessions in English or Spanish to accompany intervention content (Anthony et al., 2020). Of the 78 children randomized to *Unstuck*, 58 caregivers opted to attend English sessions, while 20 opted to attend Spanish sessions. While children improved equally regardless of caregiver session language ($p > .05$), caregivers who opted for Spanish sessions reported significantly higher satisfaction, perceived benefit for their child, and confidence in using program strategies ($p < .05$; Anthony et al., 2020). Their overall satisfaction approached the maximum of "very satisfied" ($M = 3.76$ on 0–4 scale).

Building on this, recent work has adapted *Unstuck* for telehealth. An asynchronous English-language version of *Unstuck* caregiver training, *e-Unstuck*, was shown to be as acceptable and effective as the in-person format (Kenworthy et al., 2023). This *Unstuck* online training was then condensed into 12 brief videos totaling 60 min, which caregivers rated as helpful, informative, and acceptable. Caregivers also showed significant pre-post improvements in strain, EF knowledge, and perceptions of child EF challenges (Smith et al., 2024). However, these telehealth *Unstuck* caregiver trainings have not been tested previously among predominantly Spanish-speaking families.

Current study

The current study is an extension of the previously described project (Smith et al., 2024) evaluating a set of brief *Unstuck* caregiver

videos. For this study, we recruited a cohort of caregivers who predominantly speak Spanish at home to review and evaluate Spanish translations of these videos. We replicated the methods and outcome variables from the original study with only minor adaptations and added a series of qualitative focus groups to capture participants' reflections and reactions in more depth, particularly given the limited number of asynchronous telehealth intervention studies with this population. Based on the results of the prior school-based trial (Anthony et al., 2020), we hypothesized that caregivers in this Spanish-language video cohort would similarly rate the videos as helpful, informative, and acceptable. We also hypothesized that, from pre- to post-viewing, caregivers would report improvements in their knowledge of EF principles and strategies, perceived caregiver strain, and their child's EF challenges, but that caregiver sense of parenting competency would remain unchanged. In addition, we anticipated that the focus groups would enhance the quantitative findings by offering insight into why participants rated the videos the way they did. Finally, we conducted exploratory analyses comparing this cohort's outcomes with those from the original English-language video cohort to better understand potential differences in video impact by language group.

Method

All study procedures received IRB approval. Participants were compensated for their review and feedback on the videos and participation in the focus groups, receiving comparable compensation to those in the English video cohort (up to \$205).

Procedures

See Smith and colleagues (2024) for a full review of the iterative process used to develop the original set of videos. In short, the *Unstuck* authorship team distilled the core components and strategies of the *e-Unstuck* modules into 12 videos, each lasting three to four minutes. Of the 12 videos, six were animations and six were interviews with *Unstuck* authors or families who had participated in the program. The videos included content such as an overview of executive function (EF) and its importance, coping strategies for overwhelming feelings, and the development of common language and self-regulatory phrases. The video creation process involved collaboration with a Community Advisory Board (CAB) consisting of neurodivergent self-advocates, caregivers of neurodivergent children, a diversity, equity, and inclusion consultant, and a paid video production team with expertise in health education. This process also included intentional considerations to ensure cultural and socioeconomic relevance in the selection of example scenarios and visuals (i.e. depicting modest homes and families riding the bus, discussing going to the grocery store or a family barbecue), as well as simplifying language to improve accessibility for caregivers with diverse educational backgrounds and literacy levels. Scripts were revised and reviewed iteratively by the *Unstuck* authors and CAB before being sent to the video production team. After initial animation or filming, the content was returned to the development team to suggest edits, consult further with the CAB, and ultimately approve the final product.

For the previously conducted study, all 12 videos were produced in English, but only five were translated into Spanish due to budget constraints. For the present study, we used a small grant to develop the other seven videos in Spanish. The process for creating Spanish videos was the same across projects. This included having the finalized versions of the English scripts professionally translated

Table 1
Project features aligned with ecological validity framework (Bernal et al., 1995).

Dimension	Definition	Project Features
Context	Environmental, socioeconomic, and cultural supports for the intervention	+ Short videos (~3–4 min) to fit into caregivers' busy schedules + No cost to families; freely available on YouTube+ Visual elements (e.g., people, homes) in animations reflect cultural and socioeconomic diversity + Compensated families for time reviewing videos
Methods	Intervention methods align with cultural values and practices	+ Used strategies based on common family routines + Allowed for shared viewing with children and extended family+ Included both animations and real families to maintain interest and increase relatability + Used plain language for wide literacy access+ Professionally translated videos and reviewed by bilingual content expert team
Language	Use of culturally appropriate and sensitive language	+ Non-country-specific Spanish to ensure broad comprehension- Challenging to translate some <i>Unstuck</i> -specific terms (e.g. "stuck") into universally understood Spanish + Focus groups facilitated by bilingual team members including Hispanic staff+ Some Hispanic families featured in videos- Limited representation of Hispanic voices in original footage
Persons	Consideration of ethnic/racial similarities and rapport between interventionist and family	+ Used activity examples with broad appeal (e.g. cookouts, soccer games)- Limited incorporation of "metaphors" specific to any one cultural group
Metaphors	Use of culturally familiar symbols, sayings, and themes	+ Topics relevant to parenting needs (e.g., meltdowns, routines)+ Families able to share videos widely with family and professionals- No caregiver-focused mental health content - Narrow age range
Content	Relevance of intervention topics and delivery for the community	+ Framed child behavior as part of disability rather than willful disobedience+ Promoted self-regulation and caregiver understanding- Some strategies contrast with traditional parenting techniques
Concepts	Values and norms embedded in the intervention align with those of the community	+ Emphasized calm, flexible children and reduced parent stress+ Encouraged collaborative rather than compliance-based parenting+ Designed to promote mindset shifts among caregivers to reduce stigma
Goals	Intervention goals reflect functional and culturally valued outcomes	

and then reviewed for content-specific language by three bilingual members of the research team who had experience delivering *Unstuck* to Spanish-speaking families. The *Unstuck* content experts aimed to keep key scripts/vocabulary in the most neutral or non-country-specific-Spanish possible. After this review, the scripts were sent back to the video production team, who made any requested changes. Animated videos and those featuring English-speaking *Unstuck* authors were dubbed over in Spanish, while videos with bilingual authors or families were re-recorded in Spanish. While the video creation process was not initially guided by any specific cultural adaptation framework or tailored to any one specific cultural group, we later conducted a post-hoc analysis using the Ecological Validity Model (EVM; Bernal et al., 1995) to examine how our process naturally aligned with its dimensions (see Table 1). This retrospective mapping provided insight into how cultural and contextual considerations were embedded throughout development, even without explicit reference to a formal model at the time.

Videos were distributed to participants in four sets of three videos each, with two weeks between sets. These distributions started as soon as participants completed their baseline measures. Videos were emailed to participants via a REDCap survey link, which first asked a content knowledge question related to the video. Once participants responded to the knowledge question, they were able to watch the video and then answer the same content knowledge question again. Participants were not required to view the current set of videos to receive the next set. However, upon receiving the next set they were encouraged to watch the videos and complete the surveys from the previous group if they had not done so already. All participants received the videos in the same order. Post-measures were sent two weeks after the final video set. The videos are now available on an open access playlist on YouTube (*Unstuck and On Target*, 2021a, 2021b).

Participants

This was a pragmatic study, intentionally designed to evaluate the real-world use of the videos among families likely to benefit from them. Therefore, inclusion was based solely on caregiver response to screening questions. To be included in the study, caregivers needed to meet the following criteria: 18 years or older, identify Spanish as their predominant (or only) language spoken at home, and no prior participation in an *Unstuck* research study (to understand impact of video content without prior exposure). Caregivers also needed to be the primary caregiver of a child who met the following criteria: 8–11 years old, experience EF challenges that interfere with daily life (based on two global EF interference questions that included a definition of EF; see further details in Measures section), receive supports in school, have a formal diagnosis or characteristics of autism and/or ADHD, and be rated by the caregiver as having verbal abilities approximately similar to their peers (i.e., able to speak in complete sentences). The child criteria matched those used in the prior video project (Smith et al., 2024) and a prior school-based comparative effectiveness trial of *Unstuck* (Anthony et al., 2020). A formal diagnosis was not required in order to promote equity for families who may have faced barriers to obtaining diagnostic evaluations. Additionally, prior research has shown no significant differences in response to the *Unstuck* intervention between children with formal diagnoses and those without (Troxel et al., 2024).

Participants were recruited via several sources, including community organizations, an academic medical center, and an existing research team interest list. A total of 67 caregivers were screened for inclusion, with 37 caregivers ultimately being found eligible, enrolling in the study, and completing baseline measures. The most common reasons for a caregiver not being eligible were due to their child not being the correct age or meeting the minimum verbal requirements for the study. All eligible caregivers chose to enroll in the study. Five participants (13.5 %) reviewed six or fewer videos and/or stopped responding to study staff, effectively constituting an informal withdrawal from the study, while the other 32 reviewed all 12 videos and completed all measures. See Fig. 1 for the CONSORT diagram. The 37 eligible caregivers were predominantly biological mothers (94.6 %) and represented 10 different countries of origin, with Mexico being the most common (56.7 %). Based on caregiver reports, no children had autism characteristics without a formal diagnosis, 73.0 % had a formal autism diagnosis, 10.8 % had ADHD characteristics but no formal diagnosis, and 54.1 % had a formal ADHD diagnosis. Thus, only three caregivers (8.1 %) reported their child had neither formal diagnosis. For comparisons with the cohort from the previously conducted study (Smith et al., 2024), we include the 87 caregivers who reviewed only English videos. Inclusion criteria were the same as in the current study, with the exception that participants were required to understand spoken and written English rather than Spanish. Fifteen bilingual caregivers from that project also viewed five Spanish-language videos available at the time, but because they received a mix of video languages, they are excluded from these comparative analyses for simplicity. See Table 2 for detailed demographic information.

Focus groups

Participants were invited to participate in focus groups within one month of completing the video reviews to gather additional, qualitative, information about acceptability and relevance of the videos. Given the dearth of research with this population, this mixed method design was critical to expand upon participant feedback to inform future work with this population. A total of four focus groups were run over Zoom with different times and days of the week to maximize participation options. All focus groups were conducted in Spanish.

The 32 participants who had viewed more than half of the videos were invited to participate in focus groups, as they were considered to have sufficient exposure to the content to provide meaningful feedback. The remaining five participants had stopped responding to research team communications and video surveys, effectively constituting an informal withdrawal from the study. Four

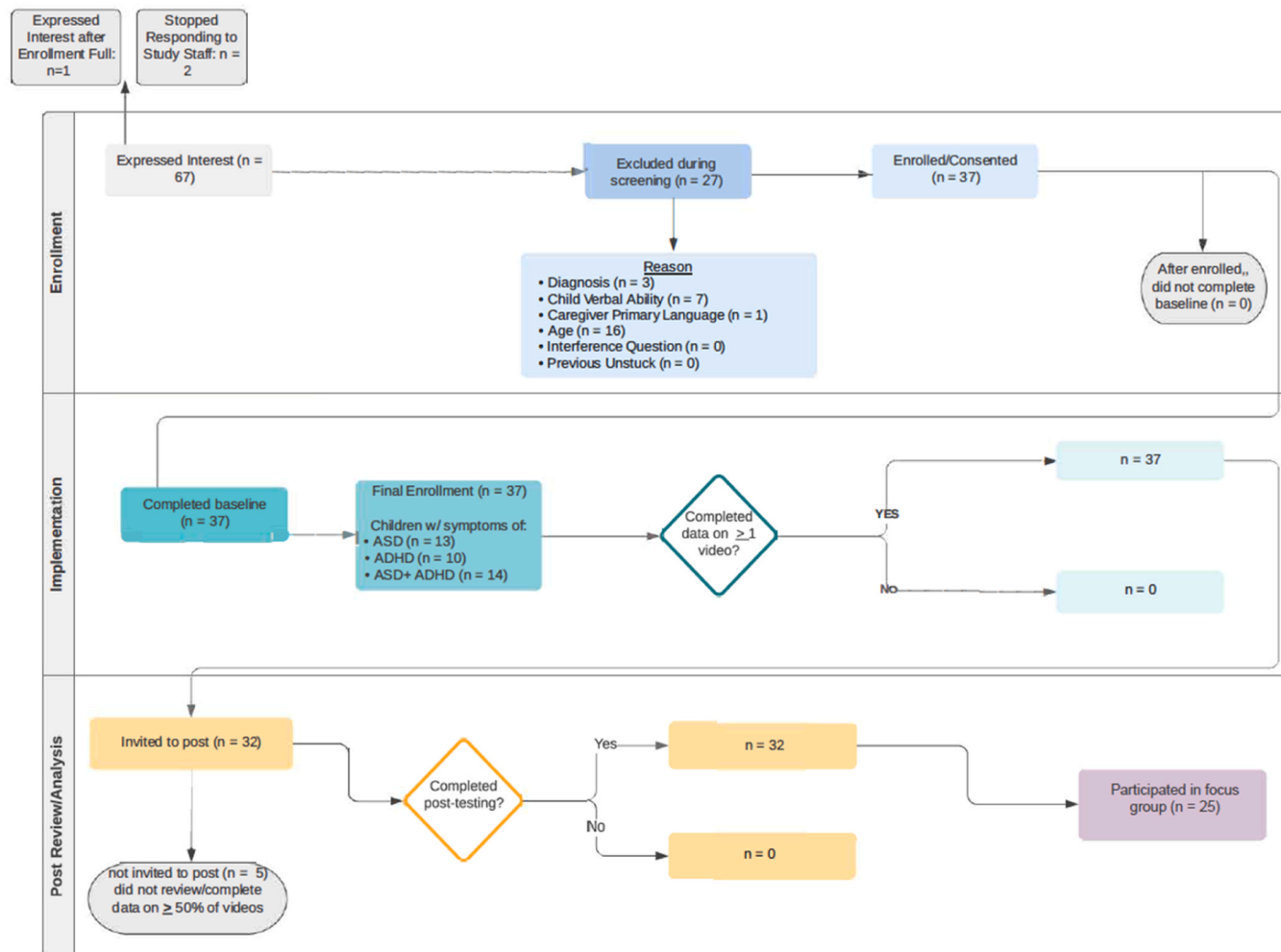


Fig. 1. CONSRT diagram of participants in Spanish video cohort.

Table 2
Sample demographics for Spanish and English video cohorts.

	Current project Spanish video cohort (<i>n</i> = 37)	Prior project English video cohort (<i>n</i> = 87)	Statistic (df)	<i>p</i> -value
Caregiver age, <i>M</i> (<i>SD</i>)	40.63 (7.85)	41.71 (6.36)	$t(122) = 0.81$	.420
Child age, <i>M</i> (<i>SD</i>)	9.50 (1.11)	9.82 (0.92)	$t(122) = 1.64$	.103
Caregiver gender, %			$\chi^2(2) = 3.16$	.206
Male	0	6.9		
Female	100	92.0		
Non-Binary	0	1.1		
Child gender, %			$\chi^2(1) = 2.41$	.121
Male	62.2	75.9		
Female	37.8	24.1		
Gender Fluid	0	0		
Caregiver-Reported Child Symptoms/Behaviors ^a , %			$\chi^2(2) = 3.03$	.220
ASD	35.1	21.8		
ADHD	27.0	40.2		
ASD and ADHD	37.8	37.9		
Child EF Frequency, <i>M</i> (<i>SD</i>) ^b	8.05 (2.11)	7.83 (1.76)	$t(122) = -0.62$	.538
Child EF Interference, <i>M</i> (<i>SD</i>) ^b	7.38 (2.28)	6.87 (1.95)	$t(122) = -1.26$	.212
Caregiver Racioethnicity, %			$\chi^2(5) = 100.64$	< .001
Hispanic/Latino(a)	97.3	5.7		
Asian	0	2.3		
Black	0	16.1		
Pacific Islander	0	1.1		
White	0	73.6		
Multiple racial/ethnic groups	2.7	1.1		
Child Racioethnicity, %			$\chi^2(6) = 98.89$	< .001
Hispanic/Latino	94.6	4.6		
Native American	0	1.1		
Asian	0	2.3		
Black	0	16.1		
Pacific Islander	0	1.1		
White	0	63.2		
Multiple racial/ethnic groups	5.4	11.5		
Caregiver Education in Years of Schooling Completed, <i>M</i> (<i>SD</i>)	12.56 (2.67)	16.85 (2.51)	$t(118) = 8.42$	< .001
Caregiver Gross Household Income (In Thousands of US dollars), %			$\chi^2(7) = 39.13$	< .001
< 30	36.1	9.2		
30–60	44.4	13.8		
61–90	2.8	17.2		
91–120	16.7	23.0		
121–150	0	11.5		
151–180	0	5.7		
181–210	0	5.7		
> 210	0	13.8		

^a As a note, symptoms and behaviors were based solely on caregiver-report. No confirmatory diagnostic measures were used.

^b Frequency and interference of child executive functioning challenges were queried through a Likert scale on a scale of 1(no/never) to 10.

evening and weekend time slots were offered, so that the 32 participants could be divided into groups of 6–10, which is the recommended size for focus groups (Morgan, 1997). A total of 25 caregivers participated across the four focus groups. Although formal data saturation was not assessed, all eligible participants were invited, and themes repeated across the later focus groups, with no new major themes emerging in the fourth group, suggesting comprehensive coverage of the topics of interest. Each group met for one hour and was facilitated by the first and second authors, who are both bilingual in Spanish/English. Focus groups used a semi-structured format to ensure topic areas were covered adequately at each group, but participants were encouraged to expand upon topics. All focus groups began with the facilitator briefly reminding participants about the main topics of the videos. The remaining time of the focus groups was divided into three sections with corresponding questions designed by the study team to explore key barriers and facilitators for these families in making use of the video content: 1) strategies/ content of the videos participants found most relevant and possible to implement with their children; 2) strategies/ content of the videos they did not like or had not tried to use with their child, and barriers to implementing strategies; and 3) generally how well they felt the video strategies/content were aligned with their individual, familial, and cultural values; whether they thought other Spanish-speaking families would find the strategies feasible and acceptable; and how the content could potentially be adapted to be more relevant to Spanish-speaking families.

Participatory methods

As described above, autism and ADHD adult self-advocates and caregivers of individuals with these conditions helped co-produce the initial set of EF videos, assisting with content creation and providing feedback to the *Unstuck* team as part of the CAB.

Measures

Caregiver satisfaction

Usefulness, Efficacy, Acceptability, and Feasibility. After each of the 12 videos, caregivers answered one question rating how informative or useful the video was on a five-point Likert scale from 1 “Not at all” to 5 “Very much” and these ratings were averaged for each video. Then, at the end of the project they rated the *overall* acceptability of video content through two questions using the same Likert scale: 1) the degree to which they found the videos valuable and 2) the degree to which they would recommend the videos to other caregivers. To measure efficacy, caregivers answered one question on the same scale, “How much did these videos change the way you think about your child’s difficulties?” Finally, feasibility of video content was assessed by asking caregivers which, if any, common *Unstuck* language words or strategies they had used with their child, and were given a list of 12 options.

Caregiver outcomes

Caregiver EF Knowledge. Caregiver EF knowledge was assessed via video-related content questions prior to and following each video. Participants answered up to 12 questions (one per video), though not all participants reviewed all videos. Examples are “EF includes all of the following...” and “A good way to help a child handle an unexpected event is...” The answers were either multiple-choice, with one correct option, or check-all-that-apply, in which participants needed to select all (and only) the correct options. Each question was coded as correct (1) or incorrect (0). For both pre- and post-video, a score was calculated by averaging the number of correct responses (out of the total answered) and multiplying by 12, representing the total number of possible questions. This yielded a scaled score with a maximum of 12.

Caregiver Strain. The Caregiver Strain Questionnaire-Short Form 7 (CSQ-SF7; Brannan et al., 2012) is a seven item self-reported questionnaire of objective and subjective internalized caregiver strain that has been translated into Spanish. Caregivers rate how difficult or impactful different situations have been over the past month as a result of their child’s behavior on a five-point Likert scale ranging from 1 “not at all” to 5 “very much.” Scores are summed for a total range of 7–35, with higher scores indicating greater strain. Previous studies have indicated that the reliability and validity of the shortened version and Spanish version are comparable to the original (Brannan et al., 2012; Domenech Rodríguez et al., 2024). In the present sample, the CSQ-7 showed good internal consistency at both pre (Cronbach’s $\alpha_{\text{Baseline}} = 0.84$) and post (Cronbach’s $\alpha_{\text{Post}} = 0.88$).

Caregiver Self-Competence. Caregiver sense of empowerment was measured through the competence subscale of the Family Empowerment Scale (FES), a brief questionnaire for assessing empowerment in families of children with disabilities (Koren et al., 1992). The eight competence items represent caregivers’ perceptions of their ability and competence as caregivers. These self-reported items use a five-point Likert scale ranging from 1 “never” to 5 “very often” for a total score range of 8–40, with higher scores indicating greater competence. The FES has previously demonstrated good validity and test-retest reliability (Singh et al., 1995). In the present sample, the FES competence subscale showed good internal consistency at both pre (Cronbach’s $\alpha_{\text{Baseline}} = 0.85$) and post (Cronbach’s $\alpha_{\text{Post}} = 0.85$).

Caregiver-report of child EF

Caregivers reported on their child’s observed EF difficulties through two global EF questions prior to and following review of the videos. Caregivers were first read a list of EF-related behaviors (following instructions, setting goals, staying organized, following plans, handling transitions and disappointments, thinking flexibly, and staying calm when faced with a challenge). They were then asked to rate 1) How often do your child’s difficulties in these areas interfere with everyday family life? and 2) When these types of difficulties do occur, how much do they disrupt everyday family life? Both items are rated on a 10-point Likert scale with higher scores indicating greater EF challenges in frequency and disruptiveness of EF challenges respectively. Scale anchors were one indicating never/none and 10 indicating more than once a day/greatest disruption possible. To be included in the study, caregivers were required to rate both items as a two or higher at baseline, to indicate at least some difficulties with EF impacting functioning. These items were adapted from the Flexibility Interference Scale, a scale created by the research team which was shown to be sensitive to change in a prior comparative effectiveness trial of *Unstuck* (Anthony et al., 2020).

Data analysis

This study followed a sequential quantitative → qualitative mixed methods design (Palinkas et al., 2011). The function of this design was expansion, in which qualitative findings elaborated on and provided deeper insight into the quantitative results, particularly explaining the reasons behind ratings of acceptability and perceived effectiveness.

Quantitative analyses within Spanish video cohort

Data were analyzed using IBM SPSS Statistics (Version 27) and R Statistical Software (Version 4.4.1). Independent samples *t*-tests or chi-squared tests were used to evaluate potential differences between those who completed more than six video reviews and those who did not on baseline characteristics such as demographic variables, caregiver strain, caregiver empowerment, and child EF. Descriptive statistics summarized acceptability outcomes. Paired *t*-tests assessed pre-post change for each continuous outcome variable: EF content knowledge, caregiver strain, caregiver self-competence, and child EF difficulties (frequency and disruptiveness). Effect sizes (Cohen’s *d*) were calculated to estimate the magnitude of change.

Qualitative data

All focus group recordings were transcribed automatically in Spanish via Zoom and then cleaned by the first author. All qualitative coding was done manually in the original Spanish. Exemplary quotes were only translated to English for this report. We used reflexive thematic analysis (Braun & Clarke, 2006; 2019) to analyze the data, drawing on Saldaña and Omasta's (2016) two-cycle coding approach. First, the first and second authors familiarized themselves with the transcripts through repeated reading, then generated initial codes line by line. A refined code list with explicit definitions was developed collaboratively, after which both authors independently applied the codes using Microsoft Excel (Meyer & Avery, 2009). We then met to compare coding, discuss discrepancies, and reach consensus. Disagreements were minimal and resolved through discussion. Consistent with reflexive approaches, the process was iterative rather than linear (Morse et al., 2002), emphasizing more flexible co-construction of meaning rather than calculating percentage agreement. Candidate themes were generated by grouping related codes, then reviewed and refined through iterative discussion, with attention to reflexivity. The first author (a bilingual non-Hispanic White psychologist) and the second author (a bilingual Hispanic research assistant), who have previously collaborated on Unstuck studies, sought to mitigate potential biases by centering participants' voices throughout the analysis and reporting. Finally, themes were defined, named, and illustrated with exemplary quotes.

Quantitative comparisons with English video cohort

Independent samples *t*-tests or chi-squared tests compared demographics differences between the Spanish and English video cohorts at baseline. A chi-square test evaluated differences among the English and Spanish video groups, in rates of completion, defined as having reviewed more than half of the videos and completed post-testing. For acceptability outcomes, feasibility (measured by the number of strategies used) was approximately normal and groups were compared with independent *t*-test; the remaining acceptability outcomes were compared between groups with Mann-Whitney *U* tests. Mixed-model analyses of covariance (ANCOVA) evaluated continuous outcome variables, with the same design: language cohort (Spanish, English), time (pre, post), and the cohort-by-time interaction. A compound symmetric covariance structure was specified to account for repeated measures, estimating between-subject variability separately from residual error. Outcomes included EF content knowledge, caregiver strain, caregiver self-competence, and child EF difficulties (frequency and disruptiveness). Finally, Spearman's correlations were calculated between each outcome at baseline and demographic variables that differed significantly by language group; significantly correlated demographics were included as covariates in the respective models.

Results

There were no differences between the completers ($n = 32$) and non-completers ($n = 5$) in the Spanish video cohort on child age, diagnosis, or EF difficulties, or on caregiver age, education, income, strain, or self-competence ($p < .05$). Mixed-methods findings related to the Spanish video cohort are displayed in full in a joint display in Table 3 and discussed below.

Table 3
Joint display for Spanish videos cohort mixed-methods results.

Quantitative Data			Qualitative Themes
	% or <i>M</i> (<i>SD</i>) out of 5		
% participants who watched all videos	86.5 %		<u>Value and relevance of video content</u>
Average usefulness per video	4.68 (0.43)		"I loved all the videos, because they reflected part of our daily situation... I personally identified with every video."
Changed view of child difficulties	4.41 (0.80)		<u>Reconceptualize EF difficulties</u>
Perceived value of videos overall	4.63 (0.71)		"We always say, he's a spoiled child, he's a crazy child, he's a disobedient child, but, well, no, right? In reality it is nothing like that, but we as adults have to educate ourselves first to be able to help them."
Likelihood to recommend	4.88 (0.42)		<u>Practical delivery modality</u>
			"I watched the videos while I was cooking and taking care of my son."
			"They're very short, very practical, and they've helped me a lot."
			"I've never had the opportunity to have videos that were in our language... And I think that helped me a lot."
EF Knowledge	<u>Pre <i>M</i></u> (<i>SD</i>) 3.54 (2.17)	<u>Post <i>M</i></u> (<i>SD</i>) 6.08 (2.49)	<u>Effect Size</u> (<i>d</i>) 1.33**
Caregiver strain	21.09 (7.09)	19.00 (6.52)	<u>Willingness to share and recommend to others</u>
Caregiver self-competence	33.66 (4.54)	33.13 (3.94)	"The fact that I was able to share them with my husband or even with my mother, gives me more strength as a mother."
Child EF difficulties (frequency)	8.09 (2.16)	5.75 (2.34)	<u>Areas for improvement, i.e. representation</u>
Child EF difficulties (disruptiveness)	7.47 (2.30)	5.19 (2.15)	"Include [more] experiences from Latino families, rather than just including videos of other families and using translation, I think it would help to feel a bit more connected to the videos."
			<u>Desired expansion to other topics</u>
			"I think you should also have made a video focusing on the parents, like how to handle a stressful situation when a child is going through something and the parent doesn't know how to handle it."

Note. * $p < .053$, ** $p < .001$.

Acceptability

Retention was high in the Spanish video cohort, with 86.5 % of enrolled caregivers reviewing more than half of the videos and completing post-testing. In fact, all 32 caregivers who reviewed more than half of the videos ultimately reviewed all 12 videos. Acceptability was also high in this cohort, with an average usefulness rating of 4.68 ($SD = 0.43$) out of 5 per video. Caregivers reported that the videos changed how they viewed their child's difficulties ($M = 4.41$, $SD = 0.80$), were valuable overall ($M = 4.63$, $SD = 0.71$), and that they were highly likely to recommend the videos to other parents ($M = 4.88$, $SD = 0.42$). For feasibility, caregivers reported using an average of 5.97 ($SD = 3.27$) of the 12 strategy options presented in the videos.

Pre-post changes

Caregiver outcomes

Caregivers in the Spanish video cohort improved in EF content knowledge from pre- to post-, $t(35) = -7.97$, $p < .001$, $d = 1.33$. Caregivers in this cohort also reported a reduction in caregiver strain, $t(31) = 2.01$, $p = .053$, $d = 0.36$, but not improvement in their sense of parenting competence $t(31) = 0.77$, $p = .443$, $d = 0.14$.

Caregiver reports of child EF difficulties

Caregivers in the Spanish video cohort reported significant reductions in the frequency of child EF difficulties, $t(31) = 4.62$, $p < .001$, $d = 0.82$, as well as the disruptiveness of child EF difficulties $t(31) = 4.60$, $p < .001$, $d = 0.81$.

Focus groups

Thematic analyses of the focus groups identified six major themes. These were: 1) enjoyed the videos overall and found the content useful and relevant; 2) changed how they conceptualized some of their child's challenges; 3) appreciated the feasible and practical delivery modality; 4) shared the videos with others; 5) noted some areas where videos could be improved; and 6) provided suggestions of other topics that could be covered in additional videos. See Fig. 2 for a summary of themes and sub-themes. Main themes are illustrated below with quotes translated into English, but the original Spanish quotes (and additional quotes) can be found in [Supplemental File 1](#). Each example quote below comes from a different participant.

Theme 1: enjoyed video content

Participants overwhelmingly spoke positively of the video content, noting that they were very useful and relevant to their daily lives. Participants described how the topics/examples felt relatable to their lived experiences of having neurodivergent children of that age. One mother shared:

"I loved all the videos, because they reflected part of our daily situation... I personally identified with every video."

And another mother expanded:

"It seemed to me that they were explained very well, because they gave examples. They gave examples, it wasn't just, ah, if you have a case of a crisis, implement this, this and this, but rather they gave you examples. So, they gave us the idea of how to implement what they were recommending."

In terms of specific content, the strategies most commonly endorsed by participants were those of managing meltdowns/disappointment and anticipating unexpected events. Strategies that participants appreciated around managing meltdowns included reducing language during moments of crisis and waiting until their child had calmed down to re-engage. Several participants across focus groups noticed that previewing what could potentially happen when they went places had helped their child mentally prepare for it and be a little better prepared to manage what occurred.

"Another thing that I implemented, which I've noticed is also working for me, is just to talk to him before a crisis could happen. That way, he already has things in his head prepared for whatever happens. He already has the solution in his head."

Finally, participants shared many examples of *Unstuck*-specific language and vocabulary that they had been trying to consistently implement with their children to help teach new skills. *Unstuck* vocabulary and strategies that were described by the greatest number of participants included Plan A/B, big deal/ little deal, compromising, and breaking things down.

Theme 2: reconceptualized child challenges

Another element of the videos that many participants reported as beneficial was around their understanding of EF and conceptualization of their child's challenges. They shared that the videos in this topic area, i.e. overview of EF in neurodivergent children and differentiation between "can't" and "won't" behaviors, helped them change how they viewed some of their child's behaviors. One mother said:

"We always say, he's a spoiled child, he's a crazy child, he's a disobedient child, but, well, no, right? In reality, it is nothing like that, but we as adults have to educate ourselves first to be able to help them."

Some of the other caregivers described actually watching the videos WITH their child so that they too could get a better

understanding of the concepts being presented. One mother said:

“I watched the videos with my child... And I told him I was learning about how to understand him. And he told me, ‘Mom, I don’t understand myself either, so I want to watch them too.’”

Theme 3: appreciated delivery modality

In addition to the video content, participants across focus groups noted that the video modality of three-to-four-minute videos about a specific topic with examples was helpful. A mother described:

“I liked that they were short, concrete, clear videos, and they helped you... I watched the videos while I was cooking and taking care of my son.”

Several participants also reported specifically appreciating that the videos were in Spanish, as they previously had limited access to these types of resources that they could understand. A few noted that the “neutral” Spanish was also helpful as they felt the videos could be understood by any native speaker regardless of their country of origin. One mother said:

“I’ve never had the opportunity to have videos that were in our language. There are many families like us who only speak Spanish, so I liked it a lot. Most studies for people like us who have children with disabilities are in English. There are very few in Spanish, and we do not have the support as parents of children with disabilities in our language. And I think that helped me a lot because it is hard to find something in Spanish.”

Finally, several participants noted that they appreciated that the videos were NOT culturally specific but rather presented strategies that were more universally ‘evidence based.’ They also felt that the strategies could be broadly applicable, i.e. not overly tied to specific diagnoses, and liked that they could use them with their neurotypical children as well. On the question of culture, one mother shared:

“Definitely, the cultural aspect is probably part of what we as parents are trying to change and grow from, and not fall into the same patterns. On the contrary, it’s like we don’t want to do what has been done all our lives in our culture, like ‘do it because I said so,’ right? That’s not going to work with our children.”

Theme 4: shared with others

Another positive aspect of the videos that was shared by many participants was that they were able to share them with other adults who they felt could benefit from the information, including their spouses and extended family members (including those living in different countries), their child’s teachers and therapists, and other Spanish-speaking caregivers of similarly aged children. One mother’s example was:

“I shared it with my mom, because she is the one who helps me with him... I’ve always wanted her to take a course on understanding autism and all that it is... and I had never had the opportunity to teach her anything and these videos helped me to get her more... involved... because my mom doesn’t speak English.”

Another mother added that she had shared them more broadly:

“I shared them at my school, and my school was fascinated by the videos, so that’s also a good thing. Additionally, at my school we have a group of moms, and we shared them among ourselves too. The more we can share them, the better, right? Because that’s where our children are, in churches and all those places, because it helps a lot, not just as a parent, right? But our whole environment in general. So yes, there are several places that need these videos.”

Theme 5: noted areas for improvement

There were also some areas noted for improvement. The biggest area for improvement noted was around increasing the representation of experiences of Hispanic families and providers. Because the original set of videos was made for a diverse group of participants, only a few of the videos specifically featured Hispanic caregivers or providers sharing their experiences. One mother offered:

“So, if the program is going to focus a bit on Latino families, maybe include experiences from Latino families, rather than just including videos of other families and using translation, I think it would help to feel a bit more connection with the videos.”

Several participants noted minor concerns around some of the vocabulary words used and the sound quality in the dubbing of the videos. Specifically, the vocabulary word for the key *Unstuck* script of “stuck” was one that several participants noted did not conceptually make as much sense in Spanish, and the terms “trapped” or “blocked” might work better. A few participants noted that the pacing or sound quality of the dubbed videos was inconsistent, with one saying:

“The only thing that seems to me is that sometimes the voices that do the translations, or the performances, are a bit robotic, right? And sometimes they’re kind of funny, so that distracted me a little bit, but nevertheless it’s not something that makes the video bad, I mean, the content of the video is perfect.”

Theme 6: provided suggestions for additional topics

Across focus groups, participants provided many suggestions for additional video topics that would be beneficial. The most

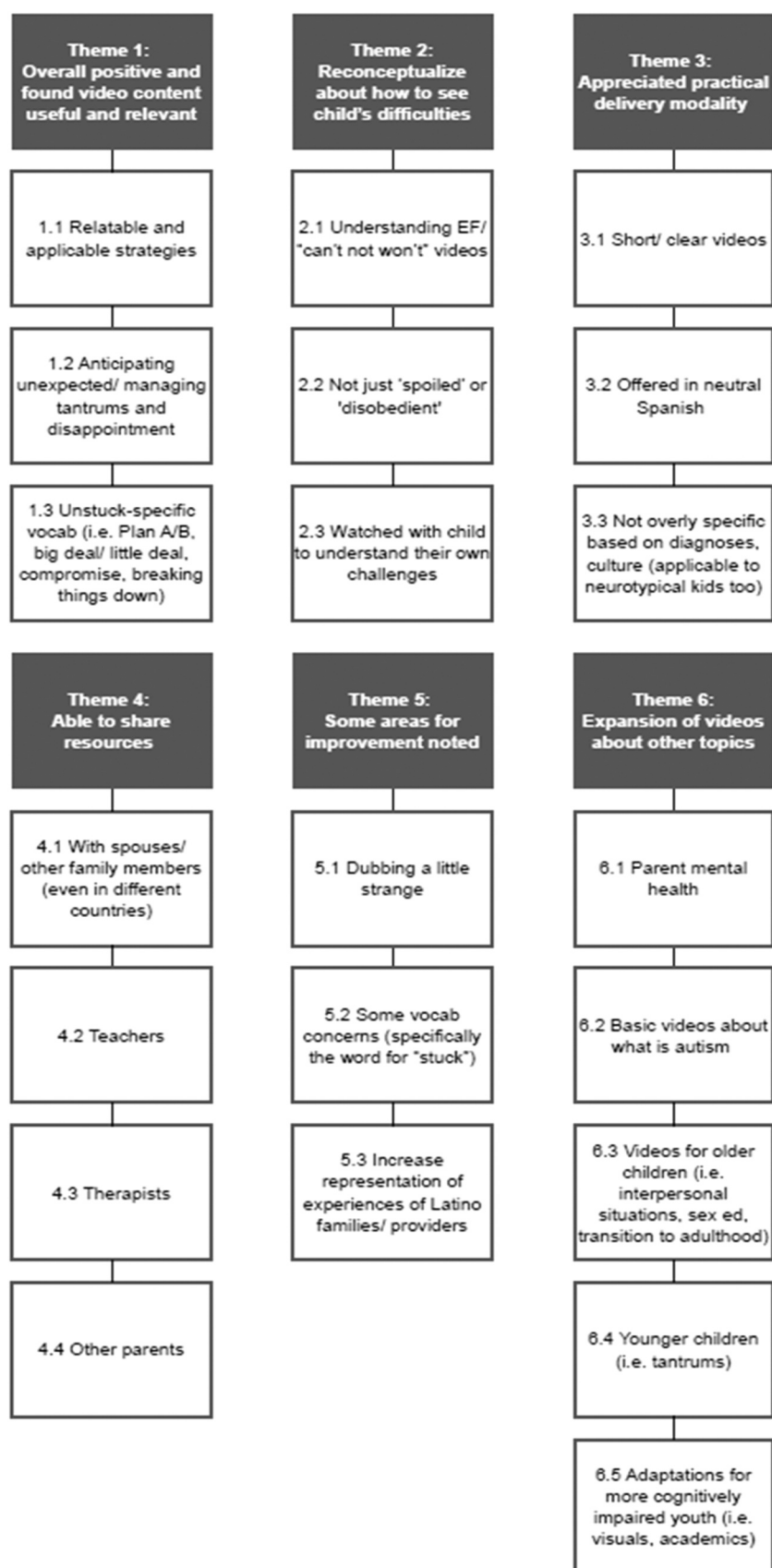


Fig. 2. Spanish video cohort focus group ($n = 25$) themes and sub-themes.

suggested video topics were around caregiver mental health, i.e. managing stress/ stigma, and helping caregivers understand how to handle different situations that arise. One mother offered:

"I think you should also have made a video focusing on the parents, like how to handle a stressful situation when a child is going through something and the parent doesn't know how to handle it."

Another mother shared:

"We need to change the mindset of the community first so that they can understand us [parents], and that we become a priority. We are the most neglected because no one worries about whether we are mentally well. We don't have anyone to turn to or ask for help."

Other suggestions made by multiple participants included more basic videos around topics such as what autism is, or videos that could adapt the EF-related content to other groups, including younger children (i.e. more focus on tantrums), older children (i.e. interpersonal situations, sexual education, transition to adulthood), or those with greater cognitive or learning difficulties.

Exploratory comparisons with English video cohort

There were several significant demographic differences between the Spanish video cohort and the prior English video cohort ($n = 87$). On average, caregivers in the Spanish video cohort had fewer years of education and lower household income than the English video cohort. The Spanish video cohort also differed from the English video cohort in race/ethnicity, as expected. See [Table 2](#) for more details. EF knowledge was the only outcome variable with which income ($r = .336$, $p < .001$) and education ($r = .520$, $p < .001$) were significantly associated at baseline, so we only evaluated each of these variables respectively as covariates in the model examining that outcome. See [Table 4](#) and [Fig. 3](#) for analysis details.

Acceptability

Proportion of caregivers in the Spanish video cohort that watched more than half of the videos and completed post-testing (86.5 %) did not differ statistically from proportion in the English cohort (75.9 %), $\chi^2(1, N = 124) = 1.77$, $p = .184$. Caregivers in the English video cohort trended to report using more strategies after viewing the videos, $t(99) = 1.83$, $p = .071$, $d = 0.39$. For all other outcomes, caregivers in the Spanish cohort provided significantly higher ratings. They rated the videos on average as more informative/useful, $U = 584.00$, $p < .001$, $r = .498$; were more likely to report a change in how they viewed their child's difficulties, $U = 548.00$, $p < .001$, $r = .393$; reported higher perceived value of the videos overall, $U = 604.50$, $p < .001$, $r = .354$; and indicated a greater likelihood to recommend the videos to others, $U = 710.50$, $p = .003$, $r = .302$.

Caregiver outcomes

Caregivers demonstrated significant improvements in EF content knowledge from pre- to post-, $F(1, 118) = 203.81$, $p < .001$, $\eta^2_p = .633$. There was also a main effect of language cohort, $F(1, 118) = 37.56$, $p < .001$, $\eta^2_p = .241$, with caregivers in the English video cohort scoring higher overall, but no significant cohort by time interaction. The language cohort effect remained highly significant even after adjusting for income and education respectively. Caregiver-reported strain decreased significantly over time, $F(1, 93) = 8.10$, $p < .01$, $\eta^2_p = .080$, with no main effect of language cohort and no interaction, indicating similar improvements across cohorts. Parenting competence did not improve over time, $F(1, 92) = 0.82$, $p > .05$, $\eta^2_p = .009$, but caregivers in the Spanish video cohort reported higher competence overall, $F(1, 92) = 13.52$, $p < .001$, $\eta^2_p = .128$, with no cohort by time interaction.

Caregiver reports of child EF difficulties

There was a significant language cohort by time interaction in the frequency of child EF difficulties from pre- to post-, $F(1, 94) = 11.70$, $p < .001$, $\eta^2_p = .111$, with caregivers in the Spanish cohort reporting greater improvements. A similar pattern was observed for disruptiveness of child EF difficulties, with a significant cohort by time interaction, $F(1, 94) = 14.47$, $p < .001$, $\eta^2_p = .133$. Again, caregivers in the Spanish video cohort reported larger decreases in perceived disruptiveness over time.

Discussion

We recruited and enrolled a cohort of predominantly Spanish-speaking Hispanic caregivers of 8–11-year-old children with autism and/or ADHD and EF difficulties to review a set of translated support videos targeting this population. This pragmatic trial was designed to evaluate the videos in a real-world context, with minimal exclusion criteria and real-world implementation. Retention in this sample was high, with 86.5 % of enrolled participants reviewing all videos and completing post-test measures. Consistent with our hypothesis, caregivers in this cohort reported high ratings of acceptability across various measures and demonstrated pre- to post-improvements in EF knowledge (very large effect), self-reported strain (small-to-medium effect), and their children's EF difficulties (large effect). These ratings of acceptability and of EF improvements in their children were even greater than those found in our prior English-language video cohort. As we have found in previous studies of online supports delivered in English ([Kenworthy et al., 2023](#); [Smith et al., 2024](#)), there were no pre- to post- improvements in self-reported feelings of competence as caregivers.

These reports of high acceptability and effectiveness of the videos were corroborated by and expanded upon during participant

focus groups, which provided insight into the reasons behind these high ratings. The majority of comments during focus groups were positive, with participant feedback highlighting *why* caregivers found the videos engaging, feasible, and impactful, including: relevant content (Theme 1), practical delivery modality (Theme 3), reframed child EF difficulties (Theme 2), and easily sharable (Theme 4). An underlying message that permeated the themes was that participants were grateful to have any resource available in their native language (Zuckerman et al., 2017), and that they often felt left out of accessing valuable resources and participating in these types of projects when Spanish-language materials were not available. While feedback was generally positive, themes related to representation (Theme 5) and desired expansion to other topics (Theme 6) highlighted areas for improvement. Caregivers requested broader expansion of video topics, including materials tailored for different child age groups and greater focus on parent mental health. They also expressed a desire for more stories and examples featuring Hispanic families to increase cultural resonance. Additionally, there were a few challenges with certain translations not fully capturing intended meanings, most notably, the key *Unstuck* concept of getting “stuck” on something was difficult to translate into universally understood Spanish, and participants felt a conceptually equivalent term might be better. These qualitative insights elaborate on and contextualize the quantitative findings, consistent with our sequential mixed-methods design, by clarifying not only how participants changed but why they believed those changes happened, and where enhancements could further strengthen the impact.

Even though the videos in the present project were not specifically tailored for Hispanic families, they were consciously designed to be broadly applicable across socioeconomic backgrounds and racial/ethnic groups, and still filled a highly desired need once translated, a finding echoed in other programs (Rollins et al., 2024). These adaptations and design choices mapped well onto the dimensions of Bernal and colleagues’ (1995) ecological validity framework, supporting the potential for cultural and contextual relevance across diverse groups. Participants did note that they were all living in the U.S. and their children participated in U.S. systems (i.e. therapies, schools, etc.), so their perspectives might differ from caregivers living in Latin American countries.

The current study’s findings of high acceptability and caregiver-reported impact in our Spanish video cohort align with prior results showing greater satisfaction among Spanish-speaking caregivers in a school-based *Unstuck* trial (Anthony et al., 2020). However, it is important to note that high acceptability and satisfaction among Spanish-speaking caregivers for intervention programs are not unique to *Unstuck*. Prior research has demonstrated similarly positive responses in programs that were explicitly and deeply culturally adapted for Spanish-speaking Hispanic families (e.g., Burke et al., 2016; Magaña et al., 2017), as well as those that were directly translated with more limited Hispanic-specific cultural adaptation (e.g., Safer-Lichtenstein et al., 2023; Martin et al., 2024; Rollins et al., 2024).

Beyond confirming the value of linguistically accessible resources, this study contributes several novel insights. Conceptually, it suggests that direct translation of well-designed, universally accessible materials can yield strong outcomes even without deep culture-specific tailoring. Theoretically, it demonstrates how Bernal’s framework can be naturally applied to pragmatic translation efforts,

Table 4
Quantitative analyses comparing Spanish ($n = 37$) and English ($n = 87$) video cohorts.

	Spanish Pre <i>M</i> (<i>SD</i>)	Post <i>M</i> (<i>SD</i>)	English Pre <i>M</i> (<i>SD</i>)	Post <i>M</i> (<i>SD</i>)	Comparisons Time	Cohort	Cohort x time
<u>Before AND after viewing each video</u>							
EF Knowledge	3.54 (2.17)	6.08 (2.49)	5.87 (1.99)	8.47 (2.08)	<i>F</i> = 203.81*** $\eta^2_p = .633$	<i>F</i> = 37.56*** $\eta^2_p = .241$	<i>F</i> = 0.03 $\eta^2_p < .001$
<u>Before AND after viewing all videos</u>							
Caregiver strain	21.09 (7.09)	19.00 (6.52)	22.25 (5.47)	21.00 (6.14)	<i>F</i> = 8.10** $\eta^2_p = .080$	<i>F</i> = 1.73 $\eta^2_p = .018$	<i>F</i> = 0.51 $\eta^2_p = .005$
Caregiver self-competence	33.66 (4.54)	33.13 (3.94)	30.42 (4.27)	30.23 (4.23)	<i>F</i> = .823 $\eta^2_p = .009$	<i>F</i> = 13.52*** $\eta^2_p = .128$	<i>F</i> = 0.18 $\eta^2_p = .002$
Child EF difficulties (frequency)	8.09 (2.16)	5.75 (2.34)	7.72 (1.76)	6.97 (1.84)	<i>F</i> = 44.10*** $\eta^2_p = .319$	<i>F</i> = 1.41 $\eta^2_p = .015$	<i>F</i> = 11.70*** η^2_p = .111
Child EF difficulties (disruptiveness)	7.47 (2.30)	5.19 (2.15)	6.75 (2.05)	6.22 (2.06)	<i>F</i> = 37.38*** $\eta^2_p = .284$	<i>F</i> = 0.16 $\eta^2_p = .002$	<i>F</i> = 14.47*** η^2_p = .133
	Spanish <i>M</i> (<i>SD</i>)		English <i>M</i> (<i>SD</i>)		Cohort Comparison		
<u>After viewing each video</u>							
Informativeness/usefulness	4.68 (0.43)		3.98 (0.68)		<i>U</i> = 584.00**, <i>r</i> = .498		
<u>After viewing all videos</u>							
Feasibility (strategies utilized)	5.97 (3.27)		7.30 (3.49)		<i>t</i> = 1.83, <i>d</i> = .391		
Changed view of child	4.41 (0.80)		3.73 (0.79)		<i>U</i> = 548.00***, <i>r</i> = .393		
Perceived value	4.63 (0.71)		4.10 (0.78)		<i>U</i> = 604.50***, <i>r</i> = .354		
Likelihood to recommend	4.88 (0.42)		4.56 (0.59)		<i>U</i> = 710.50**, <i>r</i> = .302		

Note. $***p < .001$, $**p < .01$.

Note. Positive values indicate improvement over time for knowledge and self-competence, while negative values indicate improvement for all other pre-post measures. Acceptability comparisons other than feasibility completed using nonparametric tests because not normal distributions (positively skewed).

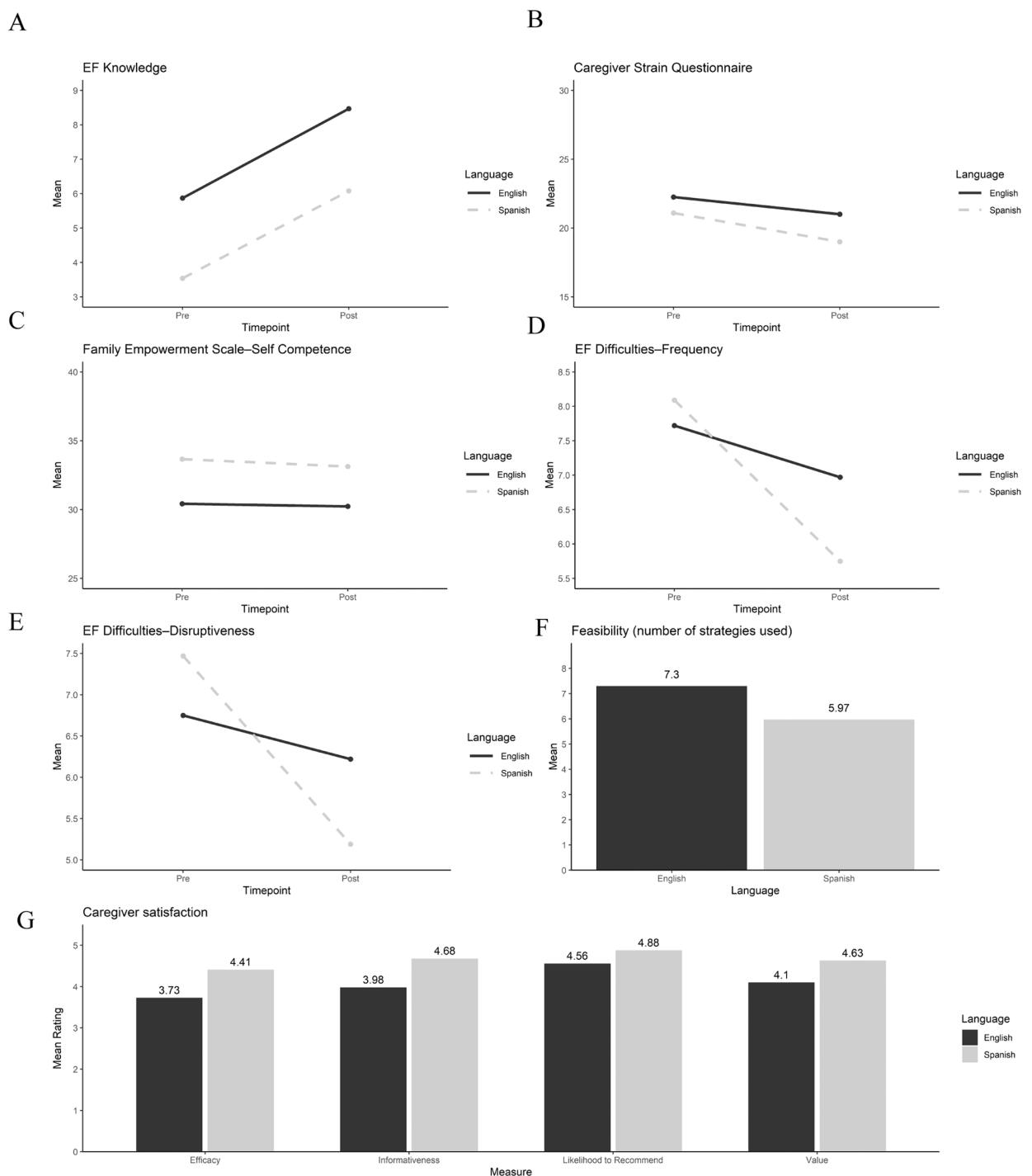


Fig. 3. Caregiver outcomes (A-C), Child Outcomes (D-E), and acceptability ratings (F-G) for Spanish ($n = 37$) and English ($n = 87$) video cohorts.

balancing scalability with cultural responsiveness. Qualitative findings provide further insight into the mechanisms underlying our observed quantitative changes, helping explain why caregivers reported improvements in EF knowledge, reductions in child EF difficulties, and high acceptability. By thoughtfully considering key factors from the ecological validity framework and translating materials into families' native languages, researchers and interventionists can make resources more accessible, relatable, and effective for diverse populations. Even short, asynchronous training videos can serve as an important resource and lead to meaningful improvements.

Limitations

This study does have several limitations. First, there was no control group used for either the prior study or the present study with the Spanish-language video cohort. Therefore, it is impossible to know if changes after watching the videos were due to the videos, although we are encouraged to see that specific improvements in knowledge of how to support children's EF occurred immediately following viewing the videos. We could not collect post-intervention or qualitative feedback from the five participants (13.5 %) in the Spanish video cohort who informally withdrew from the study and may have had less favorable opinions about the video. Additionally, all of the measures used in this study were based on caregiver reports, including their own language preference and their child's diagnoses and behaviors. For language, this means we did not evaluate the extent to which they spoke other languages, including English, which may limit the impact of the comparison to the prior cohort. For child behaviors, this means any of the changes described in this study are subject to potential desirability or confirmation biases from caregivers. There is similarly the possibility of a halo effect, wherein these caregivers overrated the impact of these videos due to simply liking them, being compensated for reviewing them, and/or appreciating that they were an option. Finally, the time periods during which the videos were watched were not the same between the Spanish and English video cohorts. The present study took place in the spring of 2024, whereas the original study took place in 2021 (Smith et al., 2024). Given the circumstances surrounding COVID-19 at that time, including school closures, families were likely experiencing additional stressors that may have impacted their ability to implement strategies from the videos.

Future directions

Based on caregiver feedback in the focus groups, an important future direction will be adding videos of Spanish-preferred caregivers sharing their experiences helping their children with EF difficulties. These videos would help reduce stigma and feelings of isolation while increasing relatability for this population. We should also consider extending the targeted age range for the videos, adding strategies to help caregivers of both younger and older children to capture common concerns at these different age and developmental levels, i.e. greater focus on meltdowns in younger children and greater focus on EF in academic and social situations for adolescents and youth with greater cognitive and language impairments. Translating the videos into additional languages beyond English and Spanish would also increase access for families. Finally, from a research standpoint, to be able to say definitively that these videos are effective, we would need to test them in a randomized trial either compared to a control group or some other type of intervention. Because these videos act as a free and easy-to-use intervention, they could also easily serve as a comparator intervention for future comparative effectiveness trials with this population.

Conclusions

Despite these limitations and areas for additional content, the positive impact found in this study suggests that providing resources in caregivers' native language, even without deep, culture-specific adaptation, can lead to meaningful outcomes. Designing interventions to be as broadly applicable or relatable as possible across socioeconomic and racial/ethnic groups, then translating them into additional languages, may offer a pragmatic and scalable alternative to more resource-intensive cultural tailoring for each individual cultural group. This approach also allows for quicker dissemination across multiple languages and contexts, while still aligning with core principles of culturally responsive practices. It can help expand access without compromising impact, particularly when materials are concise, freely available, and include clear, actionable strategies. Providing diverse language options remains essential to equitably disseminating evidence-based programs and reaching families who might otherwise lack access to traditional service systems.

CRedit authorship contribution statement

Alyssa Verbalis: Writing – review & editing, Conceptualization. **Jessica Holmes:** Writing – review & editing, Project administration, Conceptualization. **Susan Mikulich-Gilbertson:** Writing – review & editing, Methodology, Data curation. **Tennyson Dahlman:** Writing – review & editing, Visualization, Formal analysis, Data curation. **Smith Jessica:** Writing – review & editing, Conceptualization. **Laura Anthony:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Laura Campos:** Writing – review & editing, Project administration, Methodology, Formal analysis. **Lauren Kenworthy:** Writing – review & editing, Methodology, Conceptualization. **Jonathan Safer-Lichtenstein:** Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anthony Bruno:** Writing – review & editing, Methodology, Conceptualization. **Armour Anna Chelsea:** Writing – review & editing, Project administration. **Kristi Hardy:** Writing – review & editing, Conceptualization. **Allison Ratto:** Writing – review & editing, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jonathan Safer-Lichtenstein reports financial support was provided by University of Colorado Anschutz Medical Campus School of Medicine. Laura Anthony reports a relationship with University of Colorado Anschutz Medical Campus School of Medicine that includes: consulting or advisory, funding grants, and speaking and lecture fees. Lauren Kenworthy reports a relationship with Children's National Hospital that includes: consulting or advisory, funding grants, and speaking and lecture fees. If there are other

authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix B. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.reia.2025.202719](https://doi.org/10.1016/j.reia.2025.202719).

Data availability

Data will be made available on request.

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